

# Anticipating Medical Breakthroughs: An Automated Longitudinal Framework for Latent Knowledge Discovery

## Abstract

The exponential growth of biomedical literature has rendered manual Literature-Based Discovery (LBD) unfeasible, while current automated methods remain bottlenecked by human-in-the-loop dependencies and temporal inconsistencies. We present a novel, end-to-end framework for autonomous latent knowledge discovery that systematizes document analysis from raw data ingestion to actionable clinical insights. At its core, the system utilizes a domain-adaptive query generation module to mitigate lookahead bias and employs orthogonal Procrustes alignment to track the diachronic semantic trajectories of pharmacological compounds across seven decades of PubMed literature (1956–2025). To rigorously evaluate predictive utility, we introduce a formal set of metrics, including the ranking-aware Weighted LKD score, which penalizes early identifications that fail to achieve high visibility. The framework was validated through a retrospective longitudinal study on Acute Myeloid Leukemia (AML). Experimental results demonstrate that the system successfully anticipated the therapeutic potential of the compound Anagrelide in 2006—sixteen years before its official clinical report in 2022—and predicted breakthroughs for Sonidegib and Palbociclib eight years in advance. By bridging the gap between high-dimensional vector alignments and clinical interpretability via an LLM-powered contextualization layer, this work provides a scalable, reproducible foundation for accelerating the identification of non-obvious pharmacological relationships.

## CCS Concepts

• **Applied computing** → **Health informatics**; **Bioinformatics**; **Metabolomics / metabonomics**.

## Keywords

Latent Knowledge Discovery and Word Embeddings and Automated Pipeline and Document Analysis and Acute Myeloid Leukemia.

## ACM Reference Format:

. 2026. Anticipating Medical Breakthroughs: An Automated Longitudinal Framework for Latent Knowledge Discovery. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (CIKM '35)*. ACM, New York, NY, USA, 12 pages. <https://doi.org/XXXXXXX.XXXXXXX>

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CIKM '35, Rome, Italy

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ACM ISBN 978-1-4503-XXXX-X/2018/06  
<https://doi.org/XXXXXXX.XXXXXXX>

## 1 Introduction

The integration of computational tools in healthcare has changed how diagnoses and prognoses are performed. Obermeyer and Emanuel [36] emphasize that medical practice has always required professionals to perform extensive data analysis, ranging from physiological characteristics to laboratory and imaging results. However, the immense volume of available information makes manual analysis and interpretation unfeasible, highlighting the need for automated solutions to explore, confirm, or discover patterns within these data.

Official disease treatment guidelines are typically established by the World Health Organization. These procedures are generated through clinical evidence gathered over years in medical settings worldwide [23]. This process is often time-consuming and costly, as it requires validation by committees that review the development of such protocols. Furthermore, emerging diseases and variations of existing ones may demand more personalized analysis in specific clinical environments, further increasing the complexity of this task [15].

Machine Learning (ML) and Natural Language Processing (NLP) techniques offer a promising solution for extracting latent knowledge from extensive scientific datasets, such as medical corpora. This hidden knowledge consists of patterns, relationships, or insights that are not immediately apparent within a large dataset [2]. By identifying these non-obvious relationships, automated discovery enables new insights, predictions, and decisions that would not be easily achieved through traditional data analysis methods.

Tshitoyan et al. [48] proposed the use of word embeddings for latent knowledge discovery in materials science. Their approach highlights the capacity of word embeddings to contribute to the understanding of material properties and in developing new materials with potentially transformative applications. Berto et al. [4] extended the approach of Tshitoyan et al. [48] to the analysis to predict effective treatments for Acute Myeloid Leukemia (AML). The application of ML techniques to extract latent knowledge from medical texts proved promising, accurately predicting effective treatments for AML, which indicates that can accelerate the drug discovery process and provide a more robust foundation for clinical decision-making.

While the methodologies established by Tshitoyan et al. [48] and Berto et al. [4] demonstrate the robust capabilities of static word embeddings, the landscape of natural language processing offers alternative architectures that can also be applied to uncovering hidden scientific insights. Specifically, contextualized language models, such as Bidirectional Encoder Representations from Transformers (BERT) [11], represent another prominent avenue for latent knowledge discovery. Rather than using fixed word representations, these models generate dynamic embeddings from the entire surrounding text sequence, allowing them to capture shifting semantic nuances across different scientific contexts [28].

In parallel to static embedding approaches, domain-specific adaptations of these transformer frameworks have been successfully deployed to extract implicit relationships in both materials science and biomedicine. For instance, models like MatSciBERT [20] and BioBERT [28] are pre-trained on specialized corpora to identify complex entity relations, predict material properties, and map biomedical links. These contextualized frameworks provide a parallel, complementary methodology for scanning large-scale text databases, demonstrating that modern language modeling techniques, whether based on static or contextualized architectures, serve as highly versatile tools for accelerating scientific breakthroughs and therapeutic discovery.

Despite the promising results obtained with ML techniques, model generation involves a time-consuming and costly process. It requires numerous iterations with domain experts [22], extensive data analysis [25], and careful parameter tuning to maximize model performance.

The primary objective of this work is to propose an automated framework for latent knowledge discovery in medical literature. We advance the field by introducing a fully automated, domain-adaptive pipeline that eliminates the need for expert-led search queries and manual model selection. In contrast to prior studies [4, 48], which rely on manually defined seeds and limited historical validation, our proposed system operates end-to-end. It autonomously transitions from raw document analysis to optimized vector representations, producing contextualized insights rather than raw similarity scores. This automation ensures a scalable, reproducible approach to identifying potential therapeutic candidates while reducing reliance on domain-specific manual intervention. Rather than focusing exclusively on achieving predictive superiority against all existing topological and graph-based models for Literature-Based Discovery (LBD), our primary contribution centers on architectural systematization and rigorous longitudinal evaluation.

A critical advancement of this framework is the introduction of domain-adaptive query generation to mitigate *lookahead bias*. Traditional LBD methods often use static search strings that inadvertently include nomenclature or compounds that are later discovered, leading to temporal data leakage. Our framework ensures methodological integrity by dynamically adjusting queries to the historical domain’s context, ensuring that knowledge extraction remains grounded in the specific lexicon of each historical period.

Ultimately, this work is predicated upon the hypothesis that an automated, longitudinal pipeline integrating word embeddings and Large Language Models (LLMs) can accelerate therapeutic discovery by identifying non-obvious relationships within scientific corpora. The efficacy of this methodology is validated through an Acute Myeloid Leukemia (AML) case study, in which the model successfully identified promising pharmacological compounds nearly two decades before their regulatory filings.

In summary, the main contributions of this paper are as follows:

- **End-to-end discovery framework:** a fully automated pipeline for latent knowledge discovery that systematizes data ingestion, preprocessing, and model training without human-in-the-loop dependencies.
- **Domain-adaptive query generation:** a novel strategy to mitigate temporal bias by dynamically evolving search

queries, preventing data leakage and ensuring the historical validity of the discovery process.

- **Longitudinal semantic mapping:** a methodology that applies orthogonal alignment to word embeddings to track the historical semantic trajectories of chemical compounds across seven decades of biomedical literature.
- **Contextualization layer:** a bridge between geometric latent similarity and clinical interpretability, utilizing LLMs to transform raw vector relationships into actionable medical insights.
- **Empirical predictive validation:** a retrospective study on AML demonstrating the framework’s power to anticipate pharmacological breakthroughs, identifying therapeutic potentials up to 19 years before clinical recognition, thereby establishing a functional baseline for the proposed longitudinal pipeline.

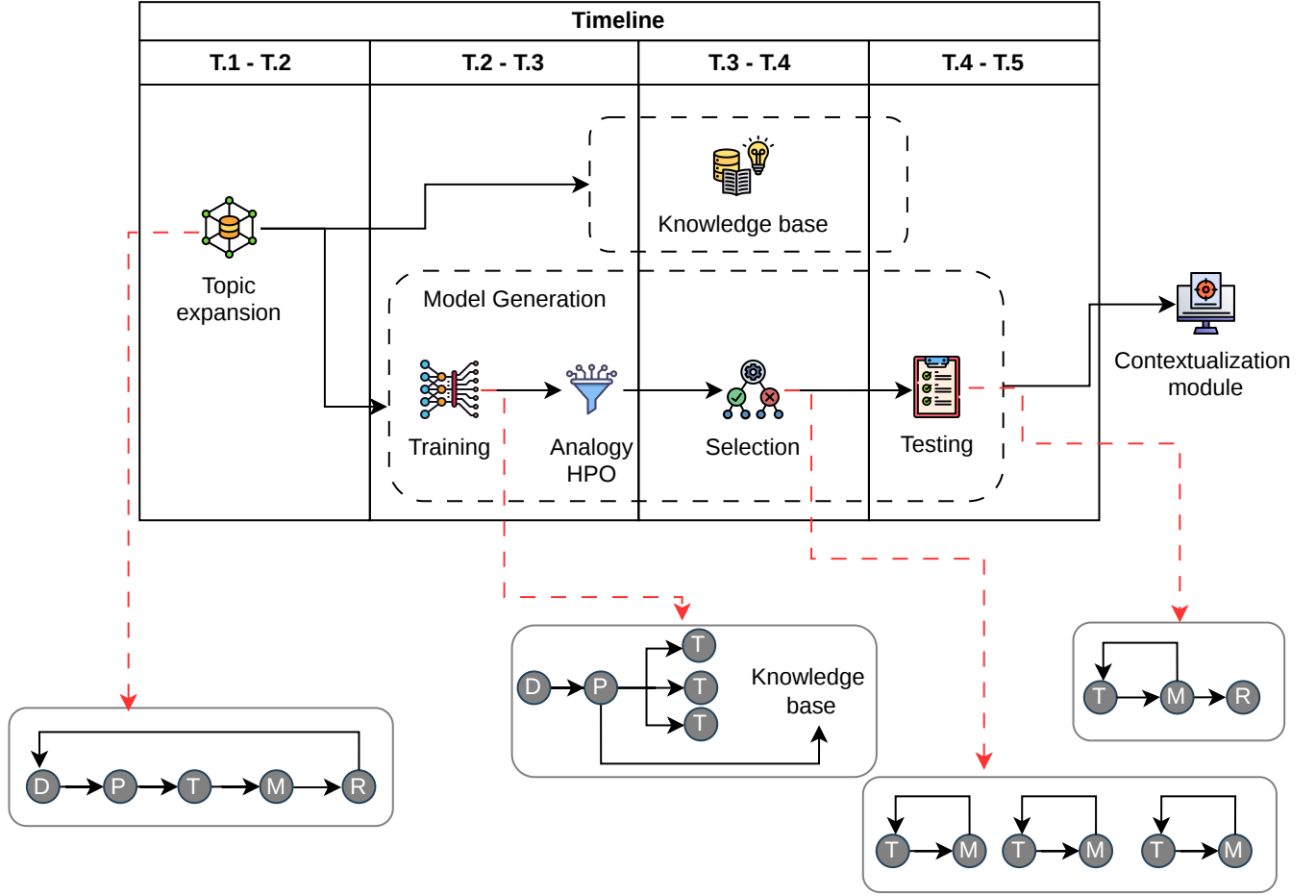
## 2 Documental analysis framework

The proposed framework, illustrated in Figure 1, introduces a methodology for latent knowledge discovery in biomedical literature. Building upon Berto et al. [4], we employ longitudinal analysis to measure the semantic distance between chemical compounds and specific medical conditions. The process begins by systematically collecting medical abstracts over a temporal window from the first recorded publication to the present. Following extraction, the raw data undergoes rigorous preprocessing to ensure input quality. For each year of study, we train multiple embedding models and optimize their configurations. The highest-performing model is then selected to conduct vector proximity analyses.

The system architecture employs an incremental growth process, annually updating the text corpus and candidate compounds to simulate diachronic state-of-the-art evolution. Within this high-dimensional latent space, vector similarity metrics quantify the proximity between the target disease and chemical compounds, prioritizing those with the highest semantic affinity. The pipeline leads to an LLM-powered contextualization module that serves as an interpretability layer. By transforming abstract vector distance metrics into actionable pharmacological hypotheses, this module elucidates latent links between compounds and pathologies, providing a robust foundation for drug repurposing.

The framework’s temporal logic is anchored to the pathology’s first recorded publication ( $T_{\text{base}}$ ), as shown in Figure 1 (T.1). Following the Topic Expansion Phase ( $T_{\text{exp}}$ , T.2), the subsequent horizon is proportionally partitioned into training (T.2–T.3), selection (T.3–T.4), and testing phases (T.4–T.5). Concurrently, an LLM-driven knowledge base is constructed (T.2–T.4) to enable historically accurate evaluation of the models’ predictive capabilities. This structured partitioning rigorously separates validation and test sets, systematically evolving raw textual data into a contextually enriched discovery model at the final stage (T.5).

The Contextualization module utilizes a domain-specific LLM to convert latent vector alignments into evidence-based narratives. Through specialized prompt engineering, the model synthesizes a compound’s mechanism of action and its theoretical link to the pathology. This provides an interpretability layer that enables rapid



**Figure 1: Overview of the proposed framework’s temporal partitioning.** T.1 ( $T_{base}$ ) marks the pathology’s origin in literature; T.2 ( $T_{exp}$ ) ends the topic expansion; T.2–T.4 represents the concurrent training with analogy resolution and LLM-based knowledge acquisition; T.5 indicates the final prospective testing and contextualization stage.

qualitative evaluation of therapeutic candidates, effectively mitigating the manual overhead of analyzing raw output.

Four modules within the framework recur across multiple temporal stages. The **Data Acquisition and Structuring** module automates the retrieval and parsing of scientific literature from APIs. It extracts metadata and core textual content from semi-structured data (e.g., XML), normalizes the information into a standard format, and aggregates records into longitudinal datasets. The **Text Normalization and Entity Resolution** module transforms raw text, filtering noise and applying medical ontologies via Named Entity Recognition (NER). It maps diverse synonyms and lexical variations to canonical identifiers, ensuring unified representations and preventing entity fragmentation.

The **Latent Discovery and Metric Quantification** module measures semantic similarity between the disease and potential therapeutic agents in latent space, filtering out non-therapeutic entities and computing cosine similarity, Euclidean distance, and similarity delta across temporal embeddings. Composite scoring and historical data are used to identify novel breakthroughs. Finally,

the **Analytical Visualization and Corpus Expansion** module synthesizes longitudinal metrics into interpretable trajectories and manages recursive corpus refinement. Orthogonal Procrustes alignment [31] ensures temporal consistency across embedding spaces. Dimensionality reduction techniques are applied to visualize candidate trajectories relative to the target disease. An iterative feedback loop expands search terms until predefined topic cardinality and thematic saturation are achieved.

## 2.1 Topic expansion (search string formulation process)

The Topic Expansion stage, situated between milestones T.1 and T.2 (Figure 1), systematically delimits the medical semantic domain while mitigating temporal anachronisms. As illustrated in the detail view of this stage in Figure 1, the process initiates with an annual data acquisition cycle (D) parameterized by the disease’s canonical name. Retrieved abstracts are refined (P) and mapped into a latent space via embedding generation (T). This geometric mapping

facilitates the discovery of unforeseen technical synonyms and cross-disciplinary concepts. By integrating knowledge from distal scientific domains, the module transcends linguistic silos to unearth obscure semantic links. These candidates are ranked by metric generation (M) based on their proximity to the target pathology and recursively integrated into the search string via the reporting module (R) for subsequent cycles. This iterative expansion ensures a comprehensive and contextually relevant corpus before the model training phase.

The framework's iterative feedback mechanism uses established treatments as semantic anchors, grounding the corpus in documented medical relationships and enabling the model to bridge disparate fields and distinguish between consolidated facts and latent discoveries. This progression continues until a predefined expansion threshold is reached, ensuring a vocabulary that is both exhaustive and representative of biomedical evolution. By processing data chronologically, the module adapts to linguistic shifts, preserving temporal integrity and validating representational stability before the selection and testing phases.

Ultimately, this process simulates the historical trajectory of scientific knowledge, mimicking the accumulation of academic evidence. By drawing insights from distal domains, the framework acts as a catalyst for latent discovery, capturing non-obvious correlations that bridge specialized silos. Grounding candidate identification solely in information published up to a specific temporal cross-section establishes a rigorous methodology for validating the model's ability to predict connections before explicit documentation. Furthermore, this feedback mechanism overcomes lexical stagnation, transforming emergent semantic associations into dynamic search parameters for subsequent discovery cycles.

## 2.2 Model training

After formulating search topics, the pipeline proceeds to the Model training phase (see the expanded view of this phase in Figure 1). In this stage, the Data collection module executes a comprehensive retrieval to aggregate all relevant abstracts published up to the current temporal cross-section. The resulting corpus is processed by the Preprocessing module to establish a normalized linguistic baseline, ensuring the extraction of stable semantic features.

The training phase is initiated by defining a hyperparameter search space for an unsupervised representation learning model. This space encompasses critical architectural parameters, such as context window size and sampling rates, to optimize the capture of latent semantic relationships. Using these configurations, the **embedding training** module generates vector representations for the initial temporal window (between moments T.2 and T.3, Figure 1).

This batch training strategy maximizes computational throughput while establishing a high-density semantic foundation for model calibration. By using a consolidated temporal window in this initial stage, the framework provides a robust baseline for validation while maintaining computational efficiency.

Finally, the **metric generation** module performs inference on candidate models. By quantifying the spatial affinity between the target pathology and pharmacological entities at the T.3 temporal

cross-section, the module ranks therapeutic candidates to initialize the subsequent selection and validation phases.

The normalized output from the Preprocessing stage (P) serves as input for the LLM-powered knowledge base construction. By providing a sanitized corpus with resolved entities, the framework ensures the language model processes data free from lexical noise and terminological ambiguity. This integration enables the LLM to synthesize biomedical evidence from a stable canonical reference.

To address the combinatorial explosion inherent in simultaneously optimizing generative architectures and their hyperparameters, which typically generates a prohibitive quadratic search space, a conditional word-pair analogy filtering mechanism is introduced during model training. The analogy dataset is divided into two primary operational domains. The first consists of grammar benchmarks derived from English, which were also employed by Tshityan et al. [48]. This linguistic subset focuses on fundamental syntactic properties, capturing relational categories such as logical antonyms (e.g., 'above' to 'below') and comparative adjective gradations (e.g., 'bad' to 'worse'). The second domain encompasses biomedical benchmarks containing relationships standardized by the Unified Medical Language System (UMLS), as detailed by Chen et al. [9]. This specialized subset tests the model's ability to map pathological conditions, clinical disorders, and surgical procedures directly to their respective anatomical structures or biological sites (e.g., 'epilepsy' to 'brain', and 'laparoscopy' to 'abdomen').

Although this filter does not reduce the time complexity required to train individual models, it effectively mitigates time and computational costs by drastically compressing the number of valid configurations from a massive quadratic scale (e.g.,  $30n^2$ ) to a highly constrained subset (e.g.,  $2n^2$ ). This systematic compression operates by evaluating models on analogy tasks and enforcing a strict selection rule: only the single best-performing hyperparameter configuration within each architectural family (e.g., the optimal Word2Vec or FastText variant) is permitted to advance. By discarding all suboptimal combinations, the mechanism ensures that exactly one representative model per architecture advances to the model selection phase. Consequently, this filtering stage transforms a complex combinatorial problem into a streamlined, linear selection process by restricting the overall number of choices rather than the training execution time itself. Crucially, this filtering is applied exclusively when comparing multiple generative architectures; in scenarios where the framework evaluates only a single architecture, the filtering phase is bypassed entirely.

This research prioritizes static word embeddings over contextualized architectures to serve as a robust proof of concept to validate the stable temporal alignment for longitudinal analysis. The requirement for invariant vector representations arises because tracking a concept's evolution over decades requires a discrete semantic centroid for each term within a given period. Static models achieve this by aggregating all occurrences of a term into a single global vector, representing the corpus collective semantic consensus at that time [21].

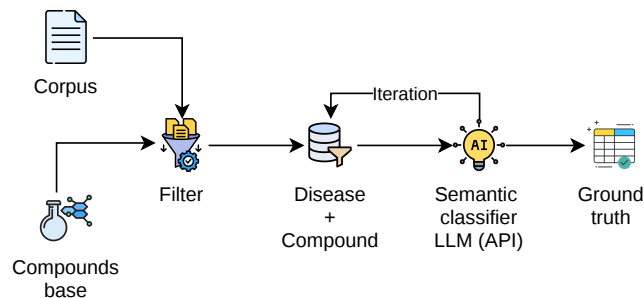
Additionally, the "one-word, one-vector" property is a functional prerequisite for Orthogonal Procrustes alignment, which requires a precise 1-to-1 mapping between different temporal vector spaces to calculate semantic shifts [46]. In contrast, the sentential variability inherent in models like BERT [11], where a single term generates

thousands of unique, context-dependent vectors, introduces a level of granularity that precludes a coherent geometric mapping of a concept’s historical trajectory.

Furthermore, the methodology is fundamentally model-agnostic with respect to the choice of static embedding. While this study implements Word2Vec [33] and FastText [7], any static architecture is suitable provided it maintains geometric stability through fixed, aggregate representations. Whether the model utilizes skip-grams or subword information (as in FastText), the resulting output remains a single point in vector space, allowing for the consistent rotation and translation necessary for longitudinal comparison.

### 2.3 LLM-Powered Knowledge Base Construction

To quantitatively benchmark the framework’s predictive capability, a retrospective ground truth is established following the workflow illustrated in Figure 2. This baseline maps each therapeutic compound to the year of its initial validation in the medical literature ( $T_{GT}^c$ ). The candidate search space is defined by applying the pharmacological filtering protocols outlined in the *Metric quantification* module. By utilizing a drug knowledge database to strictly retain small molecules in advanced clinical phases, the system ensures that the validation process is performed over the same aligned vocabulary and candidate set used during the inference stage.



**Figure 2: LLM-powered ground truth validation pipeline.** The workflow integrates the Compounds base and Corpus through a co-occurrence filter. A Semantic classifier (LLM API) then performs a chronological iteration over the identified abstracts, terminating at the first positive therapeutic validation to establish the definitive  $T_{GT}^c$ .

As described in Figure 2, the framework retrieves the subset of the corpus where both the target pathology and the compound co-occur. While computationally efficient, standard keyword-based heuristics proved insufficient for validating therapeutic relationships within these co-occurrences. Preliminary experiments indicated that rigid lexical matching failed to resolve the syntactic complexity of medical abstracts, often yielding false positives by confusing therapeutic usage with experimental toxicity or ambiguous context (e.g., “treatment induced by” vs. “treatment for”).

To resolve these semantic ambiguities, an LLM is accessed via an API-driven architecture and instantiated as a zero-shot semantic classifier. By leveraging remote inference, the framework maintains high computational scalability while evaluating each abstract against a strict prompt. The model is configured with a low-temperature

variable to ensure deterministic reproducibility, effectively eliminating stochastic variance in the classification process. The relationship identification logic adheres to the following inclusion and exclusion criteria:

- **direct therapeutic agency:** the abstract must explicitly describe the compound as an active agent resulting in clinical improvement, tumor reduction, or disease remission;
- **supportive care exclusion:** compounds mentioned solely for palliative care, symptom management, or secondary infection control are rejected;
- **causality and risk exclusion:** documents identifying the pharmacological entity as a risk factor, a cause of the pathology, or a toxic agent with no therapeutic benefit are strictly filtered out.

The process terminates at the first chronological instance in which the LLM validates the abstract as positive. The publication year of this document is recorded as the ground truth timestamp for that compound ( $T_{GT}^c$ ), serving as the definitive reference for calculating the framework’s Time-to-Discovery Gain (TDG).

### 2.4 Model selection

The model selection phase spans the temporal partition from validation onset (moment T.3) to the testing baseline (moment T.4). During this stage,  $n$  hyperparameter combinations are evaluated through an iterative, year-by-year training protocol. As illustrated in the recursive loops of the detailed view of this phase in Figure 1, the Embedding training (T) module generates successive model iterations that cumulatively integrate the historical corpus with new annual data. Simultaneously, the Metric generation (M) module computes yearly snapshots of composite heuristic scores for each pharmacological candidate.

This recursive process captures incremental semantic accumulation, identifying compounds whose vectors exhibit active migration toward the disease’s coordinates. For each candidate configuration, the system ranks the  $top-k$  compounds based on spatial proximity and convergence trajectories. These rankings serve as therapeutic recommendations for each specific temporal cross-section.

The optimal model configuration is determined by the Mean lead time metric, which quantifies the average interval between a compound’s initial recommendation and its formal validation in the LLM-augmented ground truth. By maximizing this predictive horizon, the selection process identifies the hyperparameter set most capable of unearthing therapeutic relationships at the earliest stages of their latent development.

### 2.5 Model testing

The final phase (see the detailed view of the Model Testing stage in Figure 1) uses hyperparameter optimization to perform a prospective evaluation on the most recent data. Spanning from the testing baseline (moment T.4) to the present day, the embedding training (T) and metric generation (M) modules iteratively process the accumulated corpus to generate actionable therapeutic candidates within a simulated real-world discovery environment.

To ensure interpretability, the reporting (R) module synthesizes these outputs into a comprehensive suite of visual analytics. This includes PCA projections of semantic trajectories and temporal

stability plots of heuristic scores. These quantitative metrics are then augmented by the Contextualization module, which generates qualitative descriptions for the top-ranked candidates. By fusing high-dimensional vector alignments with natural language reasoning, the framework transforms raw rankings into interpretable pharmacological hypotheses.

This integrated evidence dossier facilitates a rigorous manual validation process. While heuristic rankings reduce the candidate pool based on statistical significance, visual trajectories and LLM-synthesized rationale provide the longitudinal and biological context necessary for expert review. This multi-layered approach ensures that the final findings are clinically plausible, effectively bridging the gap between latent discovery and actionable hypothesis generation.

### 3 Evaluation Metrics for Latent Knowledge Discovery

Traditional intrinsic evaluation metrics for representation learning, such as static similarity or standard precision and recall, are insufficient for LBD, whose objective is not merely to model existing relationships, but to anticipate future scientific breakthroughs. Therefore, evaluating these models requires a framework that explicitly captures temporal dynamics, predictive relevance, and ranking quality over time.

To formalize this evaluation, we propose a novel set of metrics. Let  $C$  be the set of candidate compounds,  $t^*(c)$  denote the ground-truth discovery time (year) of compound  $c \in C$  established by the LLM-curated baseline,  $t_{\text{pred}}(c)$  be the first time the model identifies  $c$  as a relevant therapeutic candidate, and  $r_t(c)$  be the ranking position of compound  $c$  at time  $t$ .

#### 3.1 Latent Proximity and Recommendation Score

To quantify the temporal evolution of semantic similarity between the target pathology and potential therapeutic agents, the framework computes a unified proximity score.

For each compound-disease pair, the system calculates the Cosine Similarity ( $\text{Sim}_c$ ) and the Euclidean Distance ( $\text{Dist}_c$ ), alongside the year-over-year variation in similarity ( $\Delta\text{Sim}_c$ ). These spatial metrics are combined into a single ranking score:

$$\text{score}_c = \frac{\text{Sim}_c}{\text{Dist}_c} \times (1 + 10\Delta\text{Sim}_c) \quad (1)$$

This formulation strongly rewards compounds that demonstrate a semantic convergence toward the disease compared to the previous temporal window. Conversely, it penalizes entities that diverge or remain distant in the absolute latent space. Based on this metric, the top- $n$  compounds with the highest  $\text{score}_c$  are extracted as the primary therapeutic recommendations for that specific year, strictly filtering out previously consolidated treatments to focus exclusively on novel discoveries.

#### 3.2 Temporal Anticipation Metrics

The foundational metric of this framework is **TDG**. It measures how many years in advance the model anticipates a clinical discovery:

$$\text{TDG}(c) = t^*(c) - t_{\text{pred}}(c) \quad (2)$$

A  $\text{TDG} > 0$  indicates successful anticipation (forwardness),  $\text{TDG} = 0$  implies alignment with the literature, and  $\text{TDG} < 0$  denotes a delayed retrieval. To account for varying temporal windows across different corpora, this can be expressed as the **Normalized Discovery Gain (NDG)**, calculated as  $\text{NDG}(c) = \frac{\text{TDG}(c)}{t^*(c) - t_0}$ , where  $t_0$  is the start year of the corpus.

To evaluate the model's global predictive capability, we define the **Latent Knowledge Discovery Score (LKD)**, which computes the average anticipation strictly across the subset of successfully anticipated discoveries ( $C_{\text{detected}}$ ):

$$\text{LKD} = \frac{1}{|C_{\text{detected}}|} \sum_{c \in C_{\text{detected}}} \text{TDG}(c) \quad (3)$$

#### 3.3 Ranking-Aware and Composite Metrics

While predicting a compound years in advance is valuable, the utility of an LBD system heavily depends on its ability to surface these candidates at the top of the recommendation list. To penalize models that anticipate compounds early but relegate them to the bottom of the rankings, we introduce the **Weighted LKD Score**:

$$\text{LKD}_w = \frac{1}{|C|} \sum_{c \in C} \text{TDG}(c) \cdot \frac{1}{r_{t_{\text{pred}}}(c)} \quad (4)$$

This metric strictly rewards discoveries that are both early and highly ranked, serving as a critical differentiator between theoretical identification and practical applicability.

To provide a unified evaluation, the framework also tracks **Temporal Hit@K** (the fraction of future discoveries appearing in the top- $K$  predictions at time  $t$ ). These dimensions are unified into the **LKD-Composite Score**:

$$\text{LKD-Composite} = \alpha \cdot \overline{\text{NDG}} + \beta \cdot \overline{\text{Hit@K}} \quad (5)$$

where  $\alpha, \beta > 0, \alpha + \beta = 1$  are weighting coefficients (set to 0.8 and 0.2 respectively, for balanced evaluation).

### 4 Experimental setup

The framework was deployed on a high-performance server (Intel Xeon E5-2683, 256 GB RAM) utilizing a PySpark backend for the distributed processing of PubChem and PubMed datasets. AML served as the benchmark pathology for comparative analysis against established methodologies. The pipeline implemented the Word2Vec [33] and FastText [7] architectures across all semantic representation stages.

A binary search algorithm identified 1956 as the baseline year ( $T_{\text{base}}$ ) for AML. Using hyperparameters from Berto et al. [4], the module generates word embeddings and computes the most relevant compounds for the search string. While early document density (1956–1972) was insufficient for stable embeddings, the corpus achieved saturation by 1973. The automated protocol finalized a 10-term search string by 1974.

A density analysis revealed that the 1956–1972 period was too sparse for stable embeddings. However, the corpus reached saturation in 1973, allowing the automated protocol to identify cytarabine and 6-mercaptopurine. By 1974, the expansion reached its cardinality threshold with a 10-term search string. Notably, this autonomously generated set converged with the expert-curated topics utilized in the specialized AML study by Berto et al. [4], validating

the framework’s ability to reconstruct a domain-specific knowledge base without human intervention. The final search string is described below:

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(“acute myeloid leukemia”[Title/Abstract] OR “acute myeloid leukemia”[MeSH Terms]) OR (“6-mercaptopurine”[Title/Abstract] OR “6-mercaptopurine”[MeSH Terms]) OR (“cytarabine”[Title/Abstract] OR “cytarabine”[MeSH Terms]) OR (“daunorubicin”[Title/Abstract] OR “daunorubicin”[MeSH Terms]) OR (“doxorubicin”[Title/Abstract] OR “doxorubicin”[MeSH Terms]) OR (“methotrexate”[Title/Abstract] OR “methotrexate”[MeSH Terms]) OR (“prednisolone”[Title/Abstract] OR “prednisolone”[MeSH Terms]) OR (“thioguanine”[Title/Abstract] OR “thioguanine”[MeSH Terms]) OR (“vincristine”[Title/Abstract] OR “vincristine”[MeSH Terms])
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Using the expanded parameters, the system retrieved 245,437 unique abstracts up to December 2025. Based on a 60:20:20 ratio, the timeline was partitioned into topic expansion (1956–1974), model training (1975–2005), selection (2006–2017), and testing (2018–2025).

The Preprocessing module utilized the *scispaCy BC5CDR* NER model [29, 35] to isolate chemical entities, which were subsequently normalized against a PubChem-derived synonym database [26]. This created a structured vocabulary essential for temporal discovery analysis and stable embedding generation.

The retrospective ground truth ( $T_{GT}^c$ ) for therapeutic discovery was established using the *Qwen 2.5 14B Instruct* model [39]. To validate these automated timestamps, a rigorous manual audit was conducted on 20 FDA (Food and Drug Administration) <sup>1</sup> approved compounds for AML (e.g., *cytarabine*, *venetoclax*, and *revumenib*). This validation process involved a longitudinal scrutiny of all scientific abstracts published before the LLM-reported dates. The primary objective was to ensure the model’s discriminative capability in distinguishing definitive reports of clinical efficacy from mentions of supportive care, palliative use, or early-phase trials that are inconclusive. This granular verification step ensures that the temporal benchmarks used for lead-time analysis represent genuine milestones in the pharmacological literature.

To explore the high-dimensional configuration space of the embedding models, Latin Hypercube Sampling (LHS) was implemented as the primary hyperparameter optimization strategy. LHS is a multidimensional, quasi-random sampling technique that partitions each variable’s range into equal-probability intervals. This guarantees that each parameter dimension is uniformly sampled precisely once without redundant clustering [32]. This approach is highly consistent with the operational requirements of the framework, as it ensures comprehensive, evenly distributed coverage of the search space while minimizing the number of experimental trials [47]. Using LHS, the system samples a diverse set of candidate configurations for multiple architectural paradigms, specifically Word2Vec and FastText. For each configuration, as well as a predefined baseline model established in prior literature, an initial static embedding

space is trained using the accumulated corpus up to the validation onset (2005).

To mitigate the combinatorial explosion of evaluating all LHS-generated configurations across the iterative longitudinal pipeline, we introduce a static filtering mechanism based on word-pair analogies. This step evaluates the baseline semantic fidelity of each model trained up to 2005 by benchmarking against grammatical and biomedical analogy resolution tasks, ensuring that candidate models exhibit both general syntactic integrity and precise domain-specific expertise. Ultimately, only the single top-performing configuration within each architectural family advances to the next stage.

During the subsequent Selection phase, the finalists from the analogy filtering stage undergo a rigorous year-by-year training protocol to evaluate their temporal knowledge extraction capabilities. Each model’s annual recommendations are continuously benchmarked against the LLM-derived ground truth. This phase aims to definitively identify the architecture and configuration yielding the optimal TDG and overall latent knowledge discovery performance.

**Table 1: Hyperparameter configurations for best-performing candidate embedding models in analogy resolution. Both models utilize Skip-Gram architecture ( $SG = 1$ ) and 4 parallel worker threads.**

| Parameter                    | w2v_comb10 | ft_comb15 |
|------------------------------|------------|-----------|
| Dimension ( $D$ )            | 100        | 100       |
| Window ( $W$ )               | 8          | 4         |
| Min. Count ( $MC$ )          | 2          | 2         |
| Negative Samples ( $NS$ )    | 15         | 11        |
| Learning Rate ( $\alpha$ )   | 0.0354     | 0.0148    |
| Epochs ( $E$ )               | 135        | 111       |
| Neg. Sampling Exp. ( $NSE$ ) | 0.1092     | 0.0138    |
| Sub-sampling ( $S$ )         | 0.0006     | 0.0007    |
| Min. n-gram ( $min\_n$ )     | N/A        | 3         |
| Max. n-gram ( $max\_n$ )     | N/A        | 4         |

Following the identification of the optimal hyperparameter set and architecture, the framework executes a final prospective evaluation over the testing period (2018 - 2025). The resulting recommendations for the final snapshot are processed by the contextualization module. Powered by *BioMistral-7B* [27], this module generates neutral, evidence-based explanations of the chemical-pathology associations, prioritizing biochemical rationale and highlighting potential contraindications to bridge the gap between statistical discovery and clinical plausibility. This operation is executed under the prompt presented in Fig. 3.

The preprocessing pipeline and the systematic hyperparameter tuning (Table 1) establish a high-fidelity semantic space. By normalizing chemical entities and target diseases into canonical forms, the framework overcomes the lexical variability inherent in scientific literature, ensuring that the model captures implicit biochemical associations rather than mere co-occurrence noise.

Additionally, the contextualization module operates with a zero-temperature setting to mitigate stochastic hallucinations, ensuring that the identified latent relationships are corroborated by clinical

<sup>1</sup><https://www.cancer.gov/about-cancer/treatment/drugs/leukemia>



You are a medicinal chemist and clinical pharmacologist. Your task is to provide a neutral, evidence-based contextualization of a compound for a drug discovery pipeline. The pipeline has flagged <compound> as a potential candidate for <disease>. This connection may be novel/theoretical or potentially nonexistent.

CRITICAL GUIDELINES: 1. DO NOT invent clinical evidence. If no known relationship exists, describe the theoretical biochemical rationale. 2. If the compound is clearly contraindicated or toxic, explicitly state this. 3. Distinguish between 'Approved Use' and 'Theoretical Hypothesis.'

Provide a concise, evidence-focused explanation about <compound> in the context of <disease>: - Brief identification; Primary clinical/research uses; - Mechanism of action (targets/pathways); - Theoretical connection or lack of plausibility; - Plausibility assessment and safety considerations.

Respond with only the explanatory text.

**Figure 3: Prompt architecture: contextualization module.**

rationale. This experimental configuration provides the foundation for evaluating the system’s predictive power in anticipating therapeutic breakthroughs for AML.

## 5 Results and Discussion

This section presents the empirical evaluation of the proposed framework, ranging from the technical validation of the extraction modules to the qualitative analysis of latent therapeutic candidates for AML.

### 5.1 Information Extraction Performance and Language Model Evaluation

A critical contribution of this framework is the **automation of the ground-truth establishment process**, a traditionally manual task that requires extensive expert intervention. By deploying the Qwen 2.5 14B model, we replaced the laborious manual search for drug-disease associations with an autonomous pipeline. To validate this automation, a benchmark of 20 FDA-approved compounds was used for rigorous comparison.

As shown in Table 2, the LLM’s predictions for a representative subset of these compounds closely align with manually verified benchmarks. The automated process achieved high precision across the entire set, yielding a **Mean Absolute Error (MAE) of 0.75 years** and a **Root Mean Square Error (RMSE) of 1.46**. The high correlation coefficient ( $r = 0.998$ ) substantiates that our automated protocol can effectively scale the discovery of latent knowledge across massive document corpora, where manual auditing would be unfeasible.

The robustness of this autonomous curation is statistically substantiated by these error metrics; while manual curation is inherently limited by human bandwidth and subject to fatigue, our automated approach maintains expert-level precision—with an average discrepancy of less than nine months—while processing the entire pharmacological landscape of AML in a fraction of the time. This

**Table 2: Validation of the autonomous ground-truth extraction protocol. Comparison between LLM-predicted ( $T_L$ ) and expert-curated ( $T_G$ ) discovery years for AML. Global metrics (MAE: 0.75, RMSE: 1.46) across 20 compounds demonstrate high temporal precision in autonomously reconstructing historical benchmarks.**

| Compound         | $T_L$     | $T_G$     | AE                   | SE                    |
|------------------|-----------|-----------|----------------------|-----------------------|
| Cytarabine       | 1973 [16] | 1973 [16] | 0                    | 0                     |
| Daunorubicin     | 1974 [10] | 1974 [10] | 0                    | 0                     |
| Idarubicin       | 1979 [45] | 1976 [1]  | 3                    | 9                     |
| Mitoxantrone     | 1984 [37] | 1984 [37] | 0                    | 0                     |
| Cyclophosphamide | 1975 [43] | 1974 [6]  | 1                    | 1                     |
| Thioguanine      | 1975 [41] | 1973 [5]  | 2                    | 4                     |
| Doxorubicin      | 1975 [40] | 1974 [3]  | 1                    | 1                     |
| Vincristine      | 1979 [18] | 1974 [3]  | 5                    | 25                    |
| Treosulfan       | 2004 [12] | 2004 [12] | 0                    | 0                     |
| Arsenic          | 1999 [13] | 1999 [13] | 0                    | 0                     |
| Prednisone       | 1975 [8]  | 1974 [6]  | 1                    | 1                     |
| Dexamethasone    | 1996 [30] | 1996 [30] | 0                    | 0                     |
| Midostaurin      | 2004 [24] | 2003 [19] | 1                    | 1                     |
| Azacitidine      | 1976 [34] | 1975 [17] | 1                    | 1                     |
| Quizartinib      | 2009 [51] | 2009 [51] | 0                    | 0                     |
| Venetoclax       | 2014 [38] | 2014 [38] | 0                    | 0                     |
| Gilteritinib     | 2017 [14] | 2017 [14] | 0                    | 0                     |
| Enasidenib       | 2017 [49] | 2017 [49] | 0                    | 0                     |
| Ivosidenib       | 2018 [44] | 2018 [44] | 0                    | 0                     |
| Revumenib        | 2023 [42] | 2023 [42] | 0                    | 0                     |
| <b>Average</b>   | —         | —         | <b>0.75</b><br>(MAE) | <b>1.46</b><br>(RMSE) |

reliability ensures that the temporal benchmarks used to train downstream embedding models are grounded in high-fidelity therapeutic milestones.

### 5.2 Retrospective Validation and Temporal Analysis

Following the initial analogy-based filtering phase, the top-performing architectures from both paradigms, Word2Vec (w2v\_comb10) and FastText (ft\_comb15), were systematically evaluated over the Model Selection period (T.3 - T.4) using the proposed Latent Knowledge Discovery metrics. This phase aims to definitively assess which semantic representation is most effective at anticipating therapeutic milestones and prioritizing them for clinical researchers.

Table 3 summarizes the performance of both models. The empirical results demonstrate a clear superiority of the Word2Vec architecture for this specific LBD task. This is immediately evident in the global **LKD-Composite Score**, where Word2Vec achieved 0.5333, significantly outperforming FastText (0.3282).

**Table 3: Comparative evaluation of the top Word2Vec and FastText configurations using temporal discovery metrics.**

| Metric                        | Word2Vec                  | FastText                 |
|-------------------------------|---------------------------|--------------------------|
| Mean TDG (Years)              | <b>2.04</b>               | 1.40                     |
| Mean NDG                      | <b>0.046</b>              | 0.027                    |
| LKD ( $n$ detections)         | 5.27 ( $n = 22$ )         | <b>5.84</b> ( $n = 19$ ) |
| Weighted LKD                  | <b>0.3959</b>             | −0.0517                  |
| $Hit@ (5, 10, 20) \times 100$ | <b>(0.25, 0.43, 0.74)</b> | (0.11, 0.36, 0.69)       |
| LKD-Composite                 | <b>0.0379</b>             | 0.0224                   |



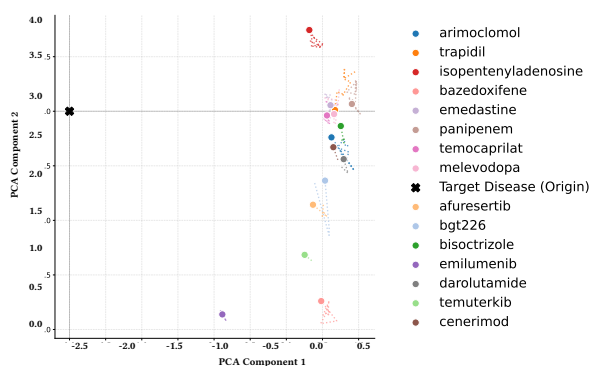
In terms of raw temporal anticipation, Word2Vec maintained a positive overall Mean TDG of 2.04 years, compared to 1.40 years for FastText. Furthermore, Word2Vec demonstrated a higher volume of successful anticipations, successfully predicting 22 compounds prior to their formal literature validation, whereas FastText anticipated 19.

However, the most critical differentiator between the two architectures lies in their ranking effectiveness, captured by the **Weighted LKD**. While FastText managed to anticipate specific compounds with remarkable lead times (such as *nabiximols*, anticipated 18 years in advance), it yielded a *negative* Weighted LKD ( $-0.0517$ ). This indicates that FastText consistently relegated its early discoveries to the bottom of the recommendation rankings. In a real-world drug discovery scenario, an insight buried low in a list is functionally invisible to a researcher.

Conversely, Word2Vec achieved a robust, positive Weighted LKD (0.3959). This demonstrates that Word2Vec not only identifies latent knowledge early but effectively pushes these highly relevant pharmacological candidates to the top of the semantic space rankings, maximizing practical utility and minimizing the researcher’s cognitive load. Ultimately, *w2v\_comb10* was selected as the definitive model for generating longitudinal trajectories and contextual insights in the subsequent testing phase.

### 5.3 Semantic Analysis and Embedding Space Visualization

The geometric evolution of the latent knowledge space is captured in Figure 4, which illustrates the non-stochastic trajectory of lead compounds toward the target disease cluster. This visualization was generated using the highest-performing model configuration identified during the testing phase (2018–2025), as detailed in the Section 5.2. By anchoring the target pathology (AML) at the origin (0, 0), the plot demonstrates a clear directional convergence—or semantic drift—of chemical entities as they accumulate scientific context over time.



**Figure 4: Aligned semantic trajectories of chemical compounds in the embedding space relative to the AML origin (0,0), generated by the optimal model configuration over the testing period (2018–2025).**

As observed in the aligned trajectories, compounds such as *Arimoclomol* and *Trapidil* exhibit a clear and consistent movement

toward the target disease. A notable case is *Bgt226*, which displays a distinct behavior compared to the tightly clustered candidates. Originating from a significantly more distant semantic region along the second principal component, *Bgt226* describes a high-magnitude “journey” across the vector space. This extensive trajectory reflects the model’s capacity to capture hidden connections from heterogeneous contexts outside the immediate disease domain.

This behavior confirms that latent knowledge is encoded geometrically long before clinical confirmation, regardless of the compound’s initial semantic positioning. The movement of these embeddings does not occur at random; rather, the “travel” toward the origin during the test period (2018–2025) serves as a visual proxy for the strengthening of the therapeutic rationale within the document corpus, allowing for the prospective identification of candidates based on their increasing proximity to the disease’s biochemical representation.

### 5.4 Case Study: Lead Compound Identification for AML

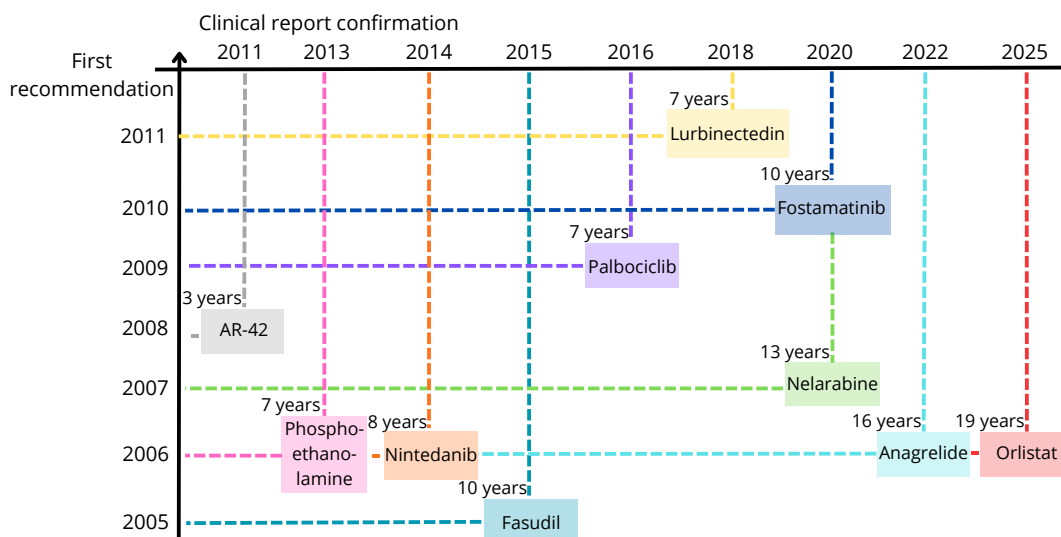
Figure 5 maps the model’s predictive alerts, plotting the first recommendation year (y-axis) against the formal clinical confirmation year (x-axis). This layout highlights the temporal gap: compounds in the lower-right region exhibit the longest latency between computational prediction and clinical validation. Demonstrating a substantial lead-time advantage, the model identified candidates like *Anagrelide* [50] and *Orlistat* as early as 2006, yielding predictive leads of 16 and 19 years, respectively, long before their clinical consolidation.

The efficacy of the framework’s temporal anticipation is most strikingly exemplified by the case of *Orlistat*. While the system issued its first recommendation in 2006, based on latent semantic proximity in the scientific corpora of that period, its therapeutic relevance for AML was only consolidated in 2025. This 19-year gap highlights the model’s ability to capture preclinical signals and complex biological patterns, potentially related to platelet modulation and interactions with the bone marrow microenvironment, decades before the formalization of clinical trials or specific reports on the target pathology.

To bridge the gap between statistical discovery and clinical relevance, the contextualization phase yielded mechanistic explanations for the top-ranked latent compounds. This module transforms geometric proximity in the vector space into interpretable pharmacological hypotheses. As illustrated in Figure 6, the system provides detailed rationales for candidates such as *Arimoclomol* and *Afuresertib*, highlighting their immunomodulatory properties and the biochemical pathways they target. This qualitative layer ensures that the notifications are not merely “opaque-box” predictions but actionable insights for drug repurposing experts.

## 6 Conclusions and future work

The existing techniques for automatically discovering new medical therapies require intensive manual intervention and domain-specific adjustments. To improve the process, this work proposed a framework for automatic latent knowledge discovery in medicine by longitudinally analyzing the semantic distance between the



**Figure 5: Predictive lead-times of identified compounds, plotting the first recommendation year against the clinical confirmation year. Trajectories highlight the temporal gap, demonstrating the model’s ability to anticipate formal clinical validations.**

**Arimoclomol:** Arimoclomol is a small molecule that belongs to the class of heat shock protein (HSP) modulators. It was developed by ArisPharma, later acquired by Astellas Pharma, and has been approved for use in Japan as an HSP modulator. The primary clinical use of arimoclomol is to treat muscle weakness associated with amyotrophic lateral sclerosis (ALS).

**Bgt226:** Bgt226, also known as 3-bromo-1-(4-chlorophenyl)-1H-indole-2-one, is a synthetic compound that belongs to the class of indole derivatives. It was first reported in 2008 by researchers at the University of Illinois who synthesized it as part of their efforts to identify novel anticancer agents.

**Trapidil:** Trapidil is a non-competitive NMDA receptor antagonist. It was developed as an anti-ischemic agent and has been used in Russia for the treatment of cerebral ischemia. The primary clinical use of trapidil is to prevent and treat ischemic brain injury, such as stroke or transient ischemic attacks. Trapidil acts by blocking the NMDA receptor, thereby reducing excitotoxicity and inflammation in the brain. This mechanism may also be relevant for treating acute myeloid leukemia (AML), a type of cancer that affects the bone marrow and blood cells.

**Tomivosertib:** Tomivosertib (also known as TAK-953 or tucatinib) is an oral, reversible, selective, and potent ATP-competitive inhibitor of the kinases PI3KCA, PI3KCD, and PI3KCG. It was developed by Novartis for solid tumors with high PI3K/AKT/mTOR pathway activity.

**Afuresertib:** Afuresertib (also known as MLN518) is a small-molecule inhibitor of the ATP-competitive kinase mTOR, which belongs to the PI3K/AKT/mTOR pathway. This pathway is involved in cell growth and proliferation, and its dysregulation has been implicated in various cancers, including acute myeloid leukemia (AML).

**Figure 6: Contextualization module output: LLM-generated rationales for high-priority candidates.**

embeddings of chemical compounds and specific medical conditions. Crucially, the foundational contribution of this research lies in the end-to-end automation of the discovery pipeline and the formalization of the evaluation metrics. We use AML as a case study, employing static representation models as a proof of concept, to validate our proposed approach by its ability to anticipate pharmacological discoveries.

This study demonstrates that automated document analysis can significantly reduce drug discovery timelines. By identifying Orlistat 19 years before its official report and predicting uses for Anagrelide and Nelarabine 16 years and 13 years in advance, respectively, the model proves its high predictive value. Ultimately, automating these insights optimizes the identification of promising compounds, serving as a powerful catalyst for accelerating medical science.

Despite the promising results, this study is subject to certain constraints. First, while the identification of search terms was automated, the selection of exactly nine compounds for the query string was determined through manual empirical observation; its mathematical optimality for broader semantic retrieval remains unvalidated. Second, the framework’s generalizability is limited by its exclusive validation of AML and its reliance on PubMed as the sole data source. Third, the current evaluation isolates static embedding architectures to validate the longitudinal Procrustes alignment. Comparative benchmarking against fundamentally different topological approaches, such as knowledge graph link prediction or traditional co-occurrence networks, falls outside the scope of this initial framework validation and represents a key avenue for future work.

In future research, we intend to expand this framework to a broader range of medical conditions to ensure its generalizability beyond AML. We aim to enhance the robustness of the framework by integrating multiple embedding algorithms and optimizing hyperparameter combinations to improve predictive accuracy. Finally, we will implement new validation metrics to better assess the clinical significance and reliability of these automated discoveries.

## Code and Data Availability

All reproducibility resources are available at [anonymized GitHub repository], including the dataset and ground truth, scripts for querying and model optimization, and pre-trained embeddings.

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